



考研竞赛凯哥 第10次周考

考试时间: 5月24日上午 08:30~10:00

讲解时间: 5月24日上午 10:00~12:00

注1: 本次周考的范围为《高数上册》《多元微分》《二重积分》(也就是数二高数部分全部内容)

注2: 周考, 每周一次, 比较频繁, 所以题量减半, 考试时间减半, 且选填题与解答题的比例随机:

月考, 每月一次, 完全按照考研的标准进行设置.

注3: 无论周考还是月考, 目的都是查漏补缺和巩固所学知识, 所以题目的难度至少持平或略高于真题的平均难度, 大家一定不要因为模考成绩不理想而灰心, 考前犯的任何错误都应该珍惜!

1. $\lim_{x \rightarrow +\infty} \left(x^{\frac{1}{x}} - 1 \right)^{\frac{1}{\ln x}}$

解:

$$\begin{aligned} I &= \lim_{x \rightarrow +\infty} \left(x^{\frac{1}{x}} - 1 \right)^{\frac{1}{\ln x}} = e^{\lim_{x \rightarrow +\infty} \frac{1}{\ln x} \ln \left(x^{\frac{1}{x}} - 1 \right)} \\ &= e^{\lim_{x \rightarrow +\infty} \frac{1}{\ln x} \cdot \frac{e^{\frac{\ln x}{x}} - 1}{\frac{\ln x}{x}} \cdot \frac{-\ln x}{x^2}} = e^{\lim_{x \rightarrow +\infty} \frac{1}{1} \cdot \left(-\frac{1}{x} / \frac{1}{x} \right)} = e^{-1} \end{aligned}$$

2. $\lim_{x \rightarrow 0} \frac{(1 + \sin 3x^2)^{\frac{1}{x^2}} - e^3}{x^n} = a (a \neq 0)$, 求 a, n .

解:

$$\begin{aligned} I &= \lim_{x \rightarrow 0} \frac{e^{\frac{1}{x^2} \ln(1 + \sin 3x^2)} - e^3}{x^n} = e^3 \lim_{x \rightarrow 0} \frac{e^{\frac{\ln(1 + \sin 3x^2)}{x^2} - 3} - 1}{x^n} \\ &= e^3 \lim_{x \rightarrow 0} \frac{\ln(1 + \sin 3x^2) - 3x^2}{x^{n+2}} \end{aligned}$$

$$\ln(1 + \sin 3x^2) = \sin 3x^2 - \frac{1}{2}(\sin 3x^2)^2 + o(x^4)$$

$$= 3x^2 - \frac{(3x^2)^3}{6} + o(x^6) - \frac{1}{2}(3x^2)^2 + o(x^4) = 3x^2 - \frac{9}{2}x^4 + o(x^4)$$

$$\ln(1 + \sin 3x^2) - 3x^2 = -\frac{9}{2}x^4 + o(x^4)$$

$$I = e^3 \lim_{x \rightarrow 0} \frac{-\frac{9}{2}x^4 + o(x^4)}{x^{n+2}}, n+2=4 \Rightarrow n=2$$

$$I = -\frac{9}{2}e^3 = a$$





3. $\lim_{n \rightarrow \infty} \frac{1}{n} \ln \frac{(3n)!}{n^{2n} \cdot n!}.$

解：

$$\begin{aligned} I &= \lim_{n \rightarrow \infty} \frac{1}{n} \ln \frac{(3n)!}{n^{2n} \cdot n!} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \ln \frac{(n+1)(n+2) \cdots (n+2n)}{n^{2n}} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^{2n} \ln \left(1 + \frac{k}{n} \right) \\ &= \int_0^2 \ln(1+x) dx \\ &= (1+x) \ln(1+x) \Big|_0^2 - \int_0^2 (1+x) \cdot \frac{1}{1+x} dx \\ &= 3 \ln 3 - \int_0^2 1 dx \\ &= 3 \ln 3 - 2 \end{aligned}$$

4. $\int \frac{1}{\sin 2x + 2 \sin x} dx$

解：

$$\begin{aligned} I &= \int \frac{1}{\sin 2x + 2 \sin x} dx \\ &= \int \frac{1}{2 \sin x \cos x + 2 \sin x} dx = \frac{1}{2} \int \frac{1}{\sin x (1 + \cos x)} dx \\ &= \frac{1}{2} \int \frac{1}{2 \sin \frac{x}{2} \cos \frac{x}{2} \cdot 2 \cos^2 \frac{x}{2}} dx = \frac{1}{4} \int \frac{1}{\sin \frac{x}{2} \cos^3 \frac{x}{2}} d\left(\frac{x}{2}\right) \\ &= \frac{1}{4} \int \frac{\sec^2 \frac{x}{2}}{\tan \frac{x}{2}} d\left(\tan \frac{x}{2}\right) = \frac{1}{4} \int \frac{1 + \tan^2 \frac{x}{2}}{\tan \frac{x}{2}} d \tan \frac{x}{2} \\ &= \frac{1}{4} \left(\ln \left| \tan \frac{x}{2} \right| + \frac{1}{2} \tan^2 \frac{x}{2} \right) + C \end{aligned}$$

5. 设 $f(x,y) = \begin{cases} \frac{x^2 y}{x^2 + y^2}, & x^2 + y^2 \neq 0 \\ 0, & x^2 + y^2 = 0 \end{cases}$, $z(t) = f(e^{2t} - 1, \arctan t)$, 则 $z'(0) = (\quad)$

- A. 0 B. $\frac{4}{5}$ C. $\frac{2}{3}$ D. $\frac{5}{4}$

解：





$$\begin{aligned} z'(0) &= \lim_{t \rightarrow 0} \frac{z(t) - z(0)}{t} \\ &= \lim_{t \rightarrow 0} \frac{(e^{2t} - 1)^2 \arctan t}{[(e^{2t} - 1)^2 + (\arctan t)^2] t} \\ &= \lim_{t \rightarrow 0} \frac{4t^2 \cdot t}{[(e^{2t} - 1)^2 + (\arctan t)^2] t} \\ &= \lim_{t \rightarrow 0} \frac{4}{\left(\frac{e^{2t} - 1}{t}\right)^2 + \left(\frac{\arctan t}{t}\right)^2} = \frac{4}{5} \end{aligned}$$

6. 证明: 当 $x > 0$ 时, $\frac{x}{\sqrt{1+x}} > \ln(1+x)$.

解:

$$\begin{aligned} \frac{x}{\sqrt{1+x}} > \ln(1+x) &\iff x^2 > (1+x)\ln^2(1+x) \\ f(x) &= (1+x)\ln^2(1+x) - x^2, \quad f(0) = 0 \\ f'(x) &= \ln^2(1+x) + 2\ln(1+x) - 2x, \quad f'(0) = 0 \\ f''(x) &= \frac{2\ln(1+x)}{1+x} + \frac{2}{1+x} - 2 = \frac{2}{1+x} [\ln(1+x) + 1 - (1+x)] = \frac{2}{1+x} [\ln(1+x) - x] < 0, \quad x > 0 \\ f(x) &= f(0) + f'(0)x + \frac{f''(\xi)}{2!} x^2 < 0 \implies (1+x)\ln^2(1+x) < x^2 \end{aligned}$$

7. 已知 $f(x)$ 为二阶可导的正值函数, $f(0) = f''(0) = 1$, $f(x)f''(x) \geq [f'(x)]^2$, 则 ()

A. $f(2) \leq e^2 \leq \sqrt{f(1)f(3)}$

B. $e^2 \leq f(2) \leq \sqrt{f(1)f(3)}$

C. $\sqrt{f(1)f(3)} \leq e^2 \leq f(2)$

D. $\sqrt{f(1)f(3)} \leq f(2) \leq e^2$

解: 选 B.

$$\begin{aligned} g(x) &= \ln f(x) \\ g'(x) &= \frac{f'(x)}{f(x)}, \quad g''(x) = \frac{f(x)f''(x) - [f'(x)]^2}{f^2(x)} \geq 0 \\ g(x) &\geq g(0) + g'(0)x, \quad g(0) = \ln f(0) = 0, \quad g'(0) = 1 \\ \ln f(x) &\geq x \implies f(x) \geq e^x \implies f(2) \geq e^2 \\ g\left(\frac{1+3}{2}\right) &\leq \frac{g(1) + g(3)}{2} \\ \ln f(2) &\leq \frac{\ln f(1) + \ln f(3)}{2} = \ln \sqrt{f(1)f(3)} \\ f(2) &\leq \sqrt{f(1)f(3)} \\ e^2 &\leq f(2) \leq \sqrt{f(1)f(3)} \end{aligned}$$





$$8. I = \int_0^1 dx \int_{x-x^3}^1 (3x^2-1)e^{y^2} dy.$$

解:

$$\begin{aligned} I &= \int_0^1 \left[\int_{x-x^3}^1 e^{y^2} dy \right] (3x^2-1) dx \\ &= (x^3-x) \int_{x-x^3}^1 e^{y^2} dy \Big|_0^1 - \int_0^1 (x^3-x) \cdot [-e^{(x-x^3)^2} (1-3x^2)] dx \\ &= 0 - \int_0^1 (x^3-x) \cdot (-1) e^{(x-x^3)^2} (1-3x^2) dx \\ &= - \int_0^1 (x-x^3) e^{(x-x^3)^2} d(x-x^3) \\ &= - \frac{1}{2} e^{(x-x^3)^2} \Big|_0^1 \\ &= - \frac{1}{2} (e^0 - e^0) = 0 \end{aligned}$$

$$9. \text{ 设 } y=g(x,z), \text{ 且 } z=z(x,y) \text{ 由方程 } f(x-z,xy)=0 \text{ 确定, 其中 } f,g \text{ 均可微, 求 } \frac{dy}{dx}, \frac{dz}{dx}.$$

解:

$$\begin{cases} y=g(x,z) \\ f(x-z,xy)=0 \end{cases} \Rightarrow \begin{cases} \frac{\partial y}{\partial x} = g_1' + g_2' \frac{\partial z}{\partial x} \quad ① \\ f_1' \left(1 - \frac{\partial z}{\partial x}\right) + f_2' \left(y + x \frac{\partial y}{\partial x}\right) = 0 \quad ② \end{cases}$$

$$f_1' - f_1' \frac{\partial z}{\partial x} + f_2' y + x f_2' \left(g_1' + g_2' \frac{\partial z}{\partial x}\right) = 0$$

$$(x g_2' f_2' - f_1') \frac{\partial z}{\partial x} = -(f_1' + y f_2' + x g_1' f_2')$$

$$\frac{\partial z}{\partial x} = \frac{f_1' + f_2' (y + x g_1')}{f_1' - x g_2' f_2'}$$

$$\frac{\partial y}{\partial x} = g_1' + g_2' \cdot \frac{f_1' + f_2' (y + x g_1')}{f_1' - x g_2' f_2'}$$





10. $\int_0^{\frac{\pi}{2}} \frac{\sqrt[3]{\tan x}}{(\sin x + \cos x)^2} dx.$

解：

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \frac{\sqrt[3]{\tan x}}{(\sin x + \cos x)^2} dx = \int_0^{\frac{\pi}{2}} \frac{\sqrt[3]{\tan x} \cdot \sec^2 x}{(\tan x + 1)^2} dx \\ &\stackrel{t = \tan x}{=} \int_0^{+\infty} \frac{\sqrt[3]{t}}{(1+t)^2} dt \\ &\stackrel{\substack{u = \sqrt[3]{t}, t = u^3 \\ dt = 3u^2 du}}{=} \int_0^{+\infty} \frac{u}{(1+u^3)^2} d(u^3 + 1) \\ &= - \int_0^{+\infty} u d\left(\frac{1}{u^3 + 1}\right) = - \frac{u}{u^3 + 1} \Big|_0^{+\infty} + \int_0^{+\infty} \frac{1}{u^3 + 1} du \\ &= \int_0^{+\infty} \frac{1}{(u+1)(u^2 - u + 1)} du \end{aligned}$$

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